Reliability Engineering and System Safety 236 (2023) 109308

Contents lists available at ScienceDirect

Reliability Engineering and System Safety

journal homepage: www.elsevier.com/locate/ress

Accident investigation and lessons not learned: AcciMap analysis of successive tailings dam collapses in Brazil

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ARTICLE INFO

Keywords:
Tailings dams
Mining industry
Accident analysis
Accident investigation
AcciMap

ABSTRACT

Tailings dam collapses, though rare, have significant human, economic and environmental impacts. In this study, we analyze two tailings dam collapses in Mariana and Brumadinho, Brazil, in 2015 and 2019 respectively, with a focus on the investigation of the first accident and the repeatability of its causal factors years later. We present an overview of tailings dams failures, followed by a literature review on these two disasters. We synthesize the findings of these reports and highlight the technical, managerial, and governmental factors which contributed to each accident, through an incremental AcciMap approach. We explore the investigation of the Mariana accident, emphasizing the "lessons not learned" which allowed similar causes to develop in Brumadinho. From these results, we propose three sets of recommendations geared towards enhancing the safety of tailings dams operation, improving the safety culture in the Brazilian regulatory environment, and improving the investigation process for dam accidents to ensure all contributing causes are uncovered and addressed. The key point in this discussion is the restructuring of the chains of accountability within dam accident response through nonpartisan inspection and investigation. We conclude with a discussion of the applications of such recommendations in reducing the risk of collapse at similar facilities.

1. Introduction

The use of dams worldwide is a primary method for storing waste from mining operations, controlling flooding, and generating renewable energy. Dams which store mining waste in the form of ground-up rock, sand, water, and other chemical reagents are known as tailings dams. Given the ease of accessibility to its rich mineral deposits, Brazil has become one of the world's leading countries in dam construction. The safety of dam use and maintenance is particularly important, considering the significant adverse consequences on local communities and the environment in case of dam failure.

On November 5th, 2015 the Fundão tailings dam in Mariana, Minas Gerais, Brazil, suffered a catastrophic collapse, flooding the communities of Bento Rodrigues and Paracatu de Baixo, killing 19 people, and releasing 43.7 million cubic meters of toxic mine tailings into the Doce river system in what became the largest environmental disaster in Brazilian history. A lengthy investigation was carried out and corrective actions were devised both to hold accountable those involved in the accident, and to prevent similar accidents from recurring. However, less than four years later, on January 25th, 2019, Dam I of the Córrego do Feijão iron mine in Brumadinho, Minas Gerais, also a tailings dam, suffered a catastrophic failure, releasing a mudflow that killed 270 people and severely contaminated the Paraopeba river basin. The basin provides much of the drinking water for the Belo Horizonte metropolitan area. Despite different immediate triggers, the two accidents shared a number of similarities at several levels, including technical, operational and organizational flaws, and there was ample time for the implementation of hazard mitigation measures in the aftermath of the first accident to prevent the second from occurring. Why this did not occur is a primary focus of the present work. With over 660 tailings dams registered in Brazil today [1], the need for a reexamination of the factors that allowed for these repeated failures to occur is timely and critical.

This work provides a comparative analysis of the two successive Mariana and Brumadinho dam disasters, and it examines how the accident investigation into the former was unsuccessful in preventing the latter from occurring. The similarities between the two disasters are identified, by analyzing the causal basis of each dam collapse using the AcciMap modeling framework [2,3]. AcciMap is a causality model which frames accidents as the result of decisions made by agents across six different levels of a safety hierarchy, including physical systems,

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https://doi.org/10.1016/j.ress.2023.109308
Received 18 January 2023; Received in revised form 4 April 2023; Accepted 10 April 2023
Available online 11 April 2023
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human operators, management, and government bodies. The Mariana and Brumadinho AcciMaps are each constructed using a novel incremental framework, which begins by identifying causal factors at the lower levels and traces them up to factors in the higher levels like regulatory oversight, safety culture, and more. This not only serves to visually outline the anatomy of each accident, it also highlights the temporal depth of causality and the diversity of agency in each case [4].

Having identified the similarities between the two disasters, the accident investigation aimed at preventing the occurrence of a second dam collapse is subsequently examined. Were there, for example, critical lessons from the first disaster that were not identified or recommendations not implemented/acted upon? An in-depth analysis of the investigation at Mariana and the recommendations issued in the aftermath of the collapse is provided. The Brumadinho collapse is then reexamined to uncover the limitations in the prior investigation. The analysis conducted in this work is a synthesis of previous technical reports, investigative reports, peer-reviewed articles and journalistic accounts of the disasters, and the authors have not conducted any additional first-hand investigation with regards to the precursors, causes and effects of the dam collapses.

The identification of the causal factors contributing to an accident plays an important role in providing a sense of comfort and closure to an accident. However, the establishment of new regulations and extensive litigation of those involved in accidents may often create a misguided sense that effective action has been taken when in fact some hazards and accident antecedents may still be uncovered, as will be shown to be the case with the successive Mariana–Brumadinho disasters. By analyzing the similarities between the two dam collapses, this work uncovers the reasons why such pathogens were allowed to remain in place despite the completion of a thorough investigation. These findings are used to provide three sets of recommendations regarding the safety of tailings dams in Brazil and worldwide. The first set is specific to tailings dams and revolves around the design and operation of such dams, the second focuses on enhancing the safety culture in Brazilian government and

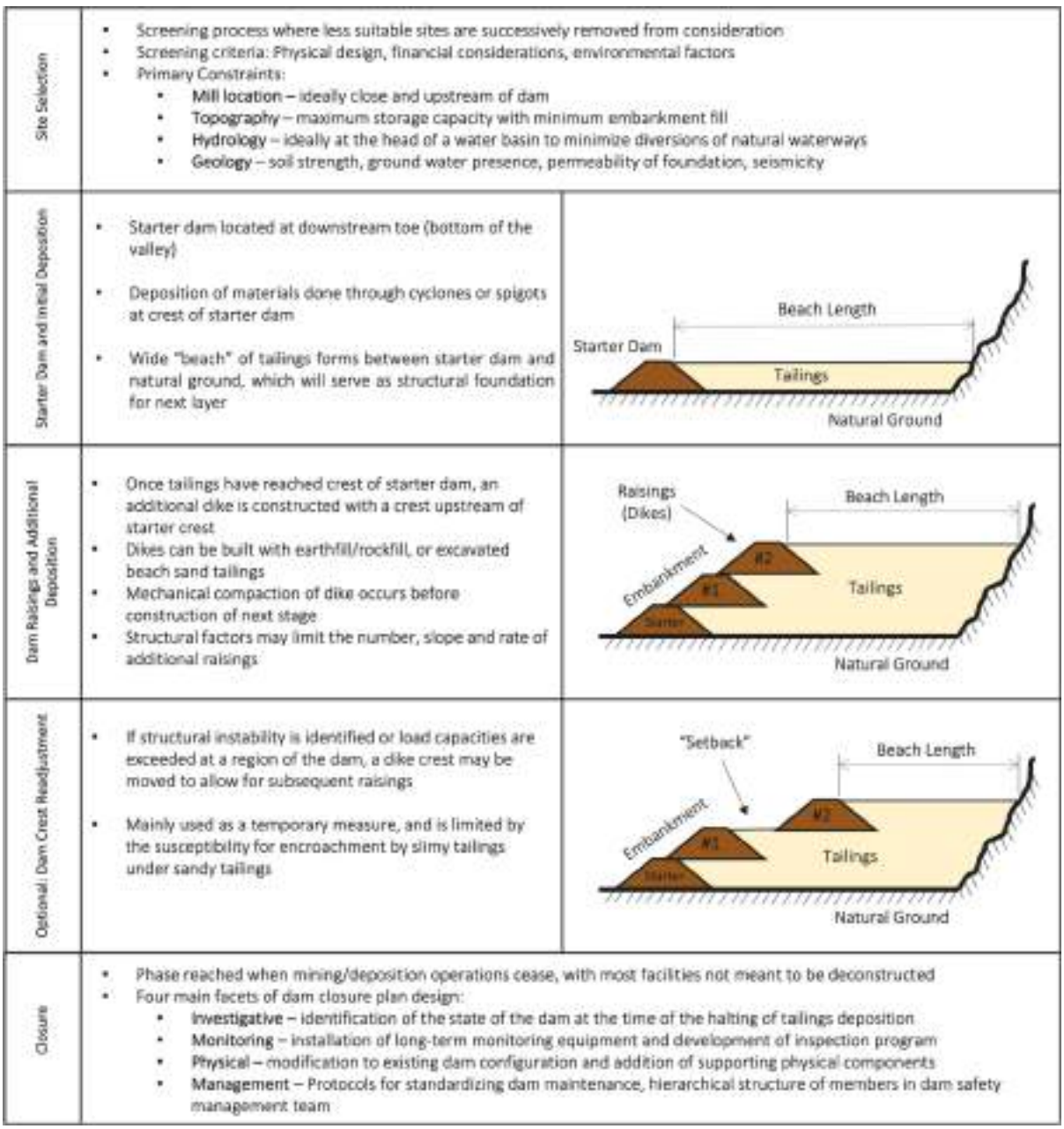

Fig. 1. Life cycle of an upstream tailings dam.

legislation, and the third focuses on broader recommendations for improving the accident investigation process itself.

The remainder of this paper is structured as follows: Section 2 outlines the basic structure and life cycle of tailings dams; Sections 3 and 4 examine and develop AcciMap models of the Mariana and Brumadinho dam disasters respectively; Section 5 analyzes the shortcomings in the Mariana accident's investigation and response, and it provides recommendations for improving tailings dam safety. Section 6 concludes the paper.

2. Background and literature review

2.1. Tailings dams: a brief introduction

The facilities at which both accidents occurred were large-scale iron mines in which the raw materials extracted from the ground are fed through crushers and screeners, to separate valuable minerals from waste. This waste, composed of ground-up rock, sand, water, and other chemical reagents, is collectively known as tailings, and is typically stored in large dams. Different from dams used for hydropower generation or flood control, tailings dams are typically earth-filled embankments whose heights are periodically increased to accommodate more tailings generated by the mining operation. The type of tailings contained in the Mariana and Brumadinho dams differed, but the dam design was largely the same. Fig. 1 depicts the typical life cycle of a tailings dam [5,6] similar to the ones operated at both facilities.

Prior to dam construction, the site is selected via a screening process during which candidate sites are identified and less suitable ones are filtered out. The screening criteria include physical, financial and environmental factors, with the major constraints on the dam's location being the region's topography, hydrology and geology, as well as the location of the mill from which the stored tailings originate (**Site Selection** layer in Fig. 1).

Once a site has been selected, a starter dam is constructed, and gradually filled using spigots or cyclones until the tailings have nearly reached the dam's rim (**Starter Dam** layer in Fig. 1). At this point, a wide "beach" of tailings will have formed behind the dam crest, on which additional dikes are constructed to "raise" the dam. These dikes are typically made using earthfill and rockfill, and are built in an *upstream* fashion, meaning that the relative position of the new crest of the dam is further upstream from the previous position [7]. This construction style results in the solidified tailings deposited at the bottom-most layer of the dam serving as a foundation for the wall raised on the layer above it (**Dam Raising** layer in Fig. 1). Such a design, though cost effective, has proven particularly risky for a variety of factors, most notably for the lack of regulations on design criteria, which allows for the use of unstable residual materials as structural foundations, and the formation of very steep embankments due to successive raisings of the dam structure [8,9].

If structural instabilities are identified, or load capacities are exceeded as a result of a dam raising, the dam's crest may be readjusted to allow for reinforcement of the compromised area (**Dam Crest Readjustment** layer in Fig. 1). This is meant as a temporary measure and is limited by the minimum beach length requirements which ensure successful drainage of the sandy tailings. Ultimately, the dam filling ends when the mining facility itself is decommissioned, or when the maximum height, volume or load capacity has been reached. As most dams are not meant to be deconstructed following closure of mining operations, closure plans are typically designed with continuous inspection and maintenance of the facility in mind. This typically includes the installation of long-term monitoring equipment, the addition of supporting physical components to the structure, and the development of protocols for standardizing dam maintenance and response to potential failures (**Closure** layer in Fig. 1). The costs accrued at this stage are recurring and may become elevated in the long term; cost-cutting at this stage often comes at the expense of safety measures at decommissioned facilities [6]. Successful and sustainable closure of tailings dams typically involves planning at the earliest stages of development, and continuous monitoring of the site such that the closing plan can be adjusted accordingly [10].

2.2. Tailings dam accidents

Tailings dam failures fall within the broad category of mining accidents explored in the work of Ismail et al. [11], and the scale of dam collapses means their impact tends to extend beyond the mine facility itself. Typical failure modes at tailings dams include collapses induced by slope instability, seismic activity, dam overtopping, seepage, foundation issues, structural issues, and mine subsidence, among others [12, 13]. Studies into such modes can be roughly split into two categories: (1) studies of the technical characteristics of the failures, and (2) studies of the overall accident scenarios that develop into and as a result of such failures.

The former category primarily focuses on risk analysis, examples of which are the work of Sun et al. [14] on overtopping risk models for earth dams, as well as the work of Du et al. [15] which focused specifically on time series modeling of the Brumadinho collapse. Morales-Torres et al. [16] proposed techniques for risk-informed management of dams in general, with special attention given to how uncertainty in the models of dams affects safety-critical decision making. Additionally, the development of methods for evaluating safety-critical parameters in dam design and operation is also relevant, and includes work such as that of Liu et al. [17] on the static and dynamic strength characteristics of tailings silty sand, and that of Andreini et al. [18] on erosion rates in embankments and earth dams. Triggers for dam failures are also diverse, and may include excessive water flow [19], exposure to unanticipated loads [20], physical deterioration of dam structures [21], among others. A single failure is typically composed of a complex interplay of technical factors, and multiple failures at different locations in dams are largely interrelated, so their combined effects are also relevant to be analyzed. This was done by Pol et al. [22] in their investigation of failure mechanism interactions in the reliability of flood control measures.

In the latter category, the literature tends to focus on monitoring and alarm systems, evacuation models, disaster management and accident investigation. The work of Alvarez et al. [23] used software to model large-scale evacuation scenarios of dam collapses, and Pisaniello et al. [24] considered the effect of dam siting on risk to downstream communities in the case of collapses or floods. Some studies also tend to examine events extending beyond the accident sequence itself, as is the case with the work of De Moura et al. [25] which examined disaster management and disaster mutual assistance specifically within Brazil, and Gonçalves Filho et al. [26] who examined investigation reports from Brazilian government bodies to determine how much is being learned from major disasters. When considering sociotechnical factors, system safety principles derived from other industries may also be leveraged for tailings dam safety. In particular, literature from the transportation industry on human evacuation [27] can be easily translated to the dam safety domain. Additionally, systematic examinations of sociotechnical, governmental, and managerial factors into accidents in different industries [28–30] may help inform analysts of what patterns to look for when studying accidents involving human interactions in their respective fields. It is a goal of this paper to combine elements of each of these analyses, to provide a comprehensive overview of the causal factors observed at Mariana and Brumadinho at several points along the accident timeline and at several layers in their safety value chains.

2.3. Previous analyses of the Mariana and Brumadinho dam disasters

In the years following the Mariana and Brumadinho dam disasters, a variety of technical reports, court documents and peer-reviewed articles have been published which explore the causes of the accidents. The

analysis conducted in this work is an amalgamation of several such reports, which are outlined in this section and referenced in the discussion of the accident profiles. In the aftermath of both disasters, expert panels were convened to determine their technical causes. The "Report on the Immediate Causes of the Failure of the Fundão Dam" [31] presents a comprehensive breakdown of the technical factors which played into the Mariana disaster, while a majority of the non-technical factors are outlined in a class action complaint [32] during the litigation procedures that followed the accident. An additional report by the Brazilian Special Secretary for Labor and Social Security (SEPRT) also examines this accident from a workplace safety perspective [33]. An equivalent "Report of the Expert Panel on the Technical Causes of the Failure of Feijão Dam I" [34] presents a technical overview of the Brumadinho disaster, while an additional panel by an independent auditing committee [35] also highlights management, cultural, and societal factors at play in the disaster. A SEPRT report [36] was also published for this accident, examining it from a workplace safety perspective.

Beyond official investigative reports, a number of peer-reviewed studies have addressed the causal factors and consequences of the dam disasters. Several such studies on the Mariana disaster place focus on the organizational and regulatory aspects [1,13,37–41] which are lacking in official reports. The works of Valencio [41] and Freitas et al. [42] conduct a critical analysis of the perceptions surrounding safety at the dam, and their impact on administrative and scientific decisions made in the leadup to the accident. A number of studies were published after the Brumadinho accident outlining its causes as well [43–47], and many drew on the similarities between the accidents to highlight repeated failures and lessons not learned [26,48–53]. Such studies tend to highlight safety hazards that were either not uncovered, not properly gauged, or inappropriately addressed in the aftermath of the first accident. The work of Botelho et al. [53] highlights the causal factors for each accident from a management, operations, maintenance, engineering and workplace environment perspective, emphasizing the similarities which should have engendered regulatory changes beyond the technical sphere. Additionally, the Brumadinho disaster also raised questions with regards to disaster response, addressed in the work of Primo et al. [54] on post-disaster decisions, as well as emergency action planning, discussed by Sampaio et al. [55] when considering the existing safety plans not properly executed during both accidents. Finally, literature exists on the changes to dam regulation following the failures [56], which allows for a deeper understanding of the long-term effects they have had on regulation and inspection procedures even outside the mining industry. The conclusions drawn from each of these studies are discussed and synthesized in the sections that follow, to present a more comprehensive examination of the accidents.

2.4. The AcciMap model

The AcciMap model was developed by Rasmussen [2] and has been widely used in safety literature in several industries including aviation, rail, maritime, and oil and gas, among others. AcciMap fundamentally revolves around the diversity of agency in accident causation; its central tenet is that accidents are the result of decisions, and actions by decision-makers across multiple levels:

> "The causal pathways of accidents are prepared as the result of decisions made by different actors, who belong to different organizations at different levels of society […]. By their various independent decisions and acts, they have shaped a causal path through the landscape along which an accident course of events sooner or later may be released." [57]

AcciMap examines the contributing factors to accidents across six hierarchical levels: the government level, the regulatory level, the company management level, the technical and operational level, the physical process level, and the equipment and surroundings level. Though not directly embedded in the model's structure, the temporal depth of causality in accidents can often be inferred from an AcciMap representation, as decisions made at different levels tend to occur at different time frames. This makes AcciMap a systemic representation of an accident, which is helpful in making generalized recommendations that tackle problems beyond their immediate manifestations in accidents. The framework has been successfully applied in several industries with different safety objectives in mind [58,59], and has been adapted in many ways to address the specific problems of the systems being analyzed [60].

3. Mariana dam disaster analysis

The Fundão tailings dam in Mariana was part of the larger Germano mining complex operated since 2008 by Samarco, a joint venture between Brazilian mining company Vale S.A. and the British-Australian BHP group. The dam housed two forms of waste: sand tailings and slime tailings. The sand tailings were stored adjacent to the dam wall structure and consisted of loose and uncompacted silt-sized particles. The slime tailings were separated from the dam structure by the sand and consisted of a low-plasticity clay which was more compressible and less permeable than the sand. To maintain the structural integrity of the dam, it was necessary that the sand tailings remain unsaturated, so that they are not susceptible to static liquefaction. This was accomplished through the installation of extensive drainage systems, as well as ensuring the separation between the slime tailings and the dam structure, so that water may drain from the sand tailings into the ground below. The surface distance of sand tailings between the dam structure and the slime tailings is termed the "beach length". A notional representation of this design is shown in Fig. 2. [7]

A schematic of the Fundão tailings dam and its immediate vicinity is depicted in Fig. 3. The outflow of water from the drainage systems in the Fundão dam joined with that of the neighboring Germano dam and was collected in the Santarém reservoir downstream. The relatively shallow Santarém dam drained out to the Santarém stream, which joined the Gualaxo do Norte River near the community of Bento Rodrigues. This river eventually flows into the Carmo and later the Doce River, the largest river in eastern Minas Gerais and in the neighboring state of Espírito Santo.

3.1. Accident timeline

This section outlines the events which immediately preceded the accident on the day it occurred, while the accident precursors which developed in the months and years leading up to the accident are discussed during the AcciMap analysis later. This accident timeline is adapted from the one presented in the official accident report [31]. On November 5th, 2015, at 1:01 and 1:06 pm, two seismic shocks of magnitude 2.1 and 2.3 were detected at the Fundão site, and they were confirmed to have resulted from planned detonations occurring at the Germano mine complex. Subsequently, between 2:13 and 2:16 pm, three additional natural earthquakes of the same magnitude were recorded, which raised no alarm as they were a frequent occurrence in the region. At the same time, work was being done for the installation of drains on the right abutment of the dam (right side of the dam crest when facing downstream). The first indication of failure came at 3:45 pm, when workers at the crest witnessed a cloud of dust drifting up the left side of the dam, followed by the formation of a crack along the left abutment. A jet of dirty water sprung from this crack, and the plateau where the dam crest readjustment had occurred immediately slid forward, causing a complete "melting" and collapse of the dam raisings above and forming a larger crack which quickly spread across the crest. The crack then progressed beyond the dam structure into the reservoir itself, at which point the central and right side of the dam disintegrated and 61% of the tailings contained were released into the valley below. The facility was not equipped with any failure detection or alert systems, so information of the collapse was conveyed over radio to the rest of the dam's workers

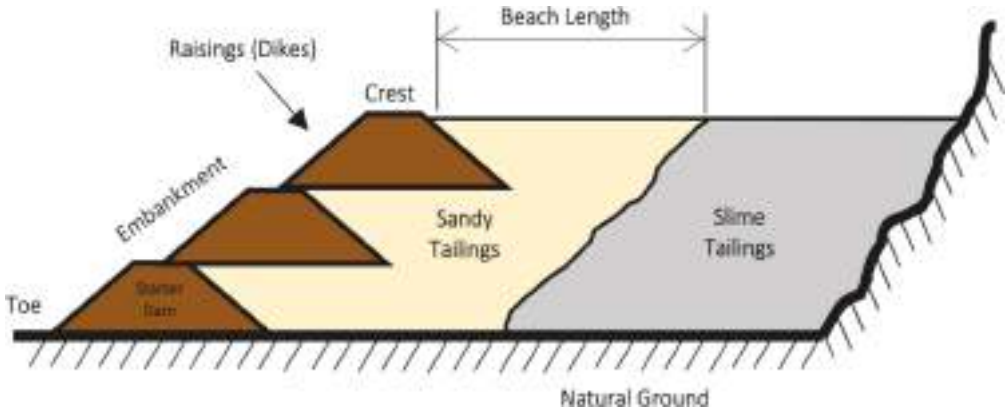

Fig. 2. Notional representation of the Fundão tailings dam in Mariana.

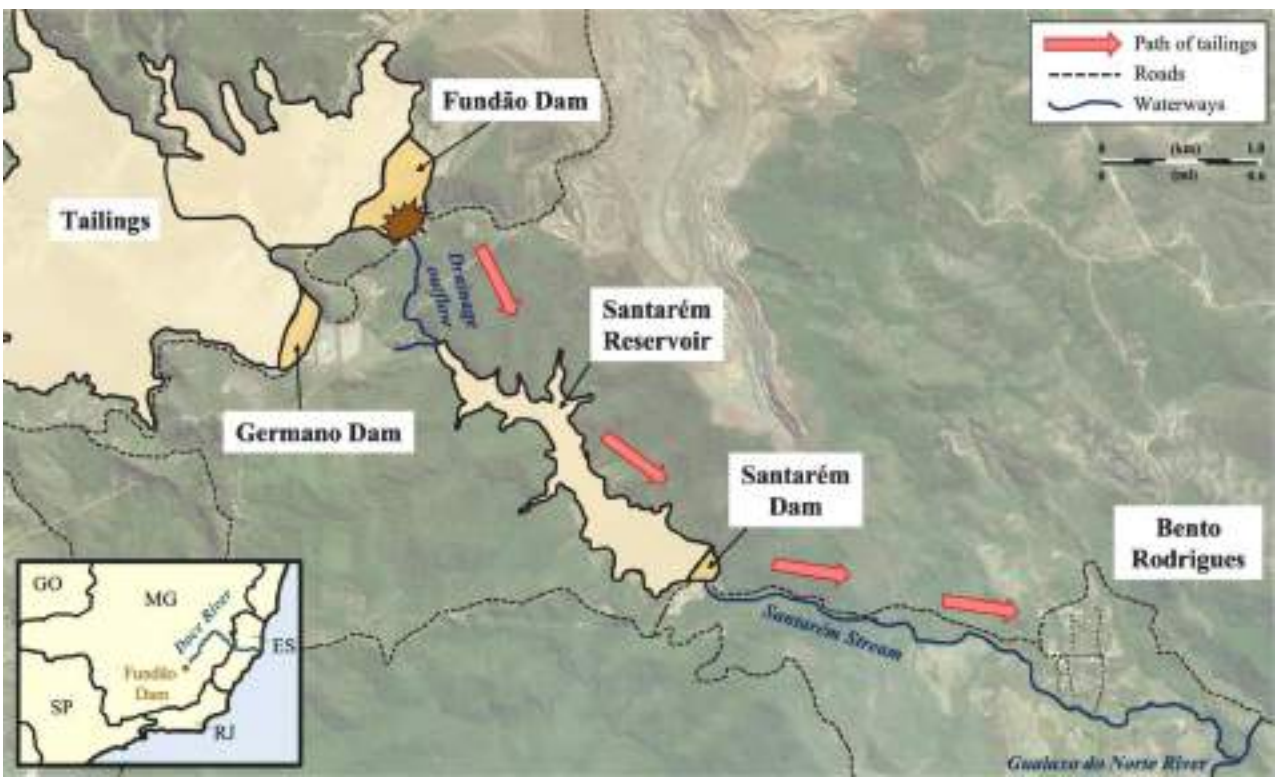

Fig. 3. Annotated image of the Fundão dam and its surrounding area prior to the accident (adapted from Google Earth).

by those who visually witnessed it. Within minutes, the wave had traveled over the Santarém dam further downstream, and in just over 10 min had covered the town of Bento Rodrigues (population 600) in tailings. 19 people lost their lives in the collapse, 13 of whom worked at the dam either for Samarco or for its subcontractors. The remaining 6 fatalities occurred in Bento Rodrigues, which had received no official notification of the collapse and was hastily evacuated when a local resident who worked at the dam rode her motorcycle to the town to warn its inhabitants of the incoming wave. Table 1 includes a condensed timeline of this sequence of events:

3.2. AcciMap analysis of the Mariana dam collapse

In this study, an AcciMap model is developed to capture the different causal factors and their connections across different levels for the Mariana and Brumadinho dam disaster. Any information featured on the AcciMap that has not yet been discussed in the accident timeline in

Table 1
Condensed Timeline of Mariana Dam Disaster.

2005	Risk of dam to Bento Rodrigues addressed, but no evacuation plan created
September-December 2008	Structural examination followed by deposition of tailings
April 2009	Large seepages detected on dam's side
2012–2014	Dam was certified as "safe", raised several times
November 5, 2015	**Day of the accident**
1:01 - 1:06 PM	Two seismic shocks caused by planned explosions
2:13 - 2:16 PM	Three natural seismic shocks Collapse of Dam
3:45 PM	Workers notified of collapse via radio Resident of Bento Rodrigues rides motorcycle to notify village of impending wave of tailings
3:49 PM	Conveyor 1300 m downstream of dam stops working
3:50 - 4:00 PM	Residents of Bento Rodrigues begin to evacuate
~3:55 PM	Wave of tailings reaches Bento Rodrigues
November 21, 2015	Tailings reach the Atlantic Ocean

Section 3.1, which displays only events that immediately precede the accident, will be discussed in detail in the sections that follow.

The novel approach to assembling the AcciMap proposed here is *incremental*, beginning with the bottom two levels, representing the physical events, equipment, conditions which immediately led up to the accident, and moving upward from there. Rather than identifying an all-encompassing set of causal factors, placing each of them within the most appropriate level, and establishing causal links post-facto, the incremental approach identifies the causal factors at one level and uses them to guide the generation of the factors in the levels above. This is particularly helpful when it comes to traceability of causality, as no causal factor is treated in isolation, and each is understood by tracing it to factors in the levels above it. It is important to note that causality may also flow from lower levels to upper levels, as decisions made by government or management may stem from actions being taken at the operational level. The incremental approach is also particularly helpful when identifying causal factors that are shared among accidents involving the same company or industry. When the incremental construction of two AcciMaps yields similar results in the upper levels despite differences at the lower levels, more effective recommendations may be generated that address a less localized set of causal factors.

Fig. 4 begins the incremental assembly of the Mariana dam disaster AcciMap by highlighting the bottom 2 levels of the model. Color-coding of boxes within AcciMap is not necessary, but may prove helpful in grouping causal factors that exhibit the same broad themes, in order to keep track of causal factors across multiple levels of the hierarchy. In this example, the blue box groups factors relating to the physical structure of the tailings dam, the green box groups factors relating to the dam's location/environment, and the orange box groups factors relating to alert systems and emergency response.

The failure of Fundão tailings Dam in Mariana is a prime example of the temporal depth of causality in accidents, as no single action taken during the day of the accident actually triggered the collapse. To understand the underlying failure mechanisms, it is necessary to analyze the dam's structure, location, and the emergency measures it had in place.

A historical overview of the Fundão tailings dam is described in [31, 33], and synthesized here. In September 2008, a structural examination of the starter dam was conducted, and the mine owner/operator Samarco obtained authorization to begin depositing tailings at the site by December. The design featured an extensive underground drainage system to extract moisture from the sand tailings, but in April 2009 large seepages were observed on the side of the dam, along with the detection of internal erosion of the sand tailings [53]. A subsequent investigation of the already certified dam revealed **flaws in the drainage system's construction**, including a section of the drain's outlet that was never completed. The impending rainy season meant that work to repair the drainage systems from the erosion was impossible at the time and so the drains, a fundamental piece of the original dam design, were sealed off. To address this problem, a new drainage system was installed and relied on a "blanket drain", which introduces a layer of very permeable material to spread out low-velocity runoff across a hill and avoid seepage zones. The system was installed above the tailings already deposited in the dam, ensuring sufficient draining of any new tailings deposited, but having no effect on the saturation of existing tailings. The saturation was compounded when the less porous **slimy tailings extruded underneath the sandy tailings**, thus trapping water in the system. Frequent additional rainy seasons further contributed to the already damaged drainage systems, which eventually compromised the structure and initiated a process of static liquefaction.

Earlier, in 2005, during the location selection process for Samarco's newest dam, the risks to the community of Bento Rodrigues were identified and highlighted in the dam's Environmental Impact Study (EIA) [61]. Three location alternatives were discussed in this report, with the final selection being the only one directly upstream of Bento Rodrigues, as well as the Santarém reservoir, potentially compounding the amount of tailings released in the event of a collapse Accordingly, this report placed the dam's risk at "Class III", the highest level, due its height, volume and location [39], but the **impacts were only determined to be "moderate"** [40]. Thus the construction of the dam at this site was allowed, with the condition that a network of continuous communication between the dam's administration and the city's leadership be established to facilitate evacuations if needed [62]. **Such a network was never established**, and simulations of a dam collapse were only held at Samarco's facility, rather than in any of the neighboring communities [32]. **No sirens had been installed at the facility or the communities** [53], and without any official notification of the dam collapse, many residents were reluctant to believe in the danger, and thus delayed their own evacuation, costing several lives [32]. This reflects an intrinsic connection between causal factors at different levels

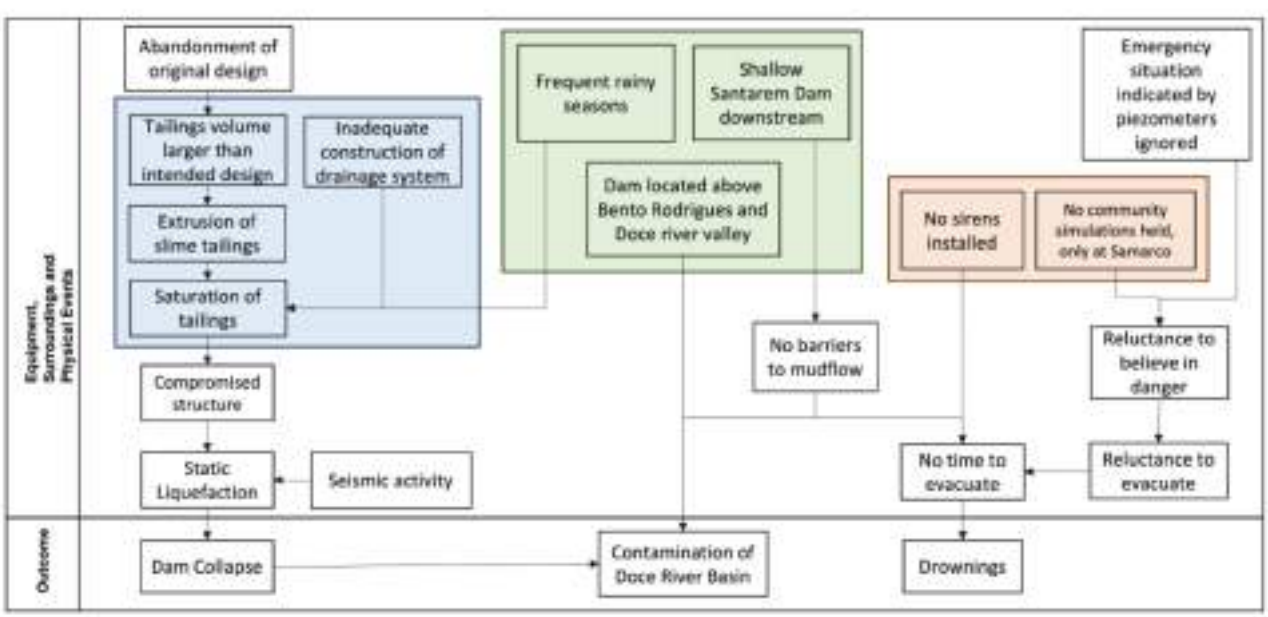

Fig. 4. Partial AcciMap for the Mariana Dam Disaster.

of defense-in-depth, as the alert systems and evacuation plans act as safety barriers to mitigate the consequences of the accident after it has already occurred, while the dam siting acts several layers above that in determining what the potential consequences of the accident might be. The need for the mitigation barriers stems directly from decisions made in the siting process, which highlights the importance of understanding each step of the dam design process in determining which accident pathogens may exist.

The next two rows in the AcciMap model, representing the technical and company management elements of the accident, are condensed in Fig. 5. The red box here captures factors relating to dam design decisions, and the purple box captures issues of company safety culture.

The inadequate construction of the dam's original drainage system can be traced back to an overall disregard for technical oversight at Samarco, driven primarily by cost reduction strategies [32]. The upstream dam design, which relied on the tailings as the dam's structural foundation, was plagued with known issues [5], yet it was still selected as the most suitable design for the facility [32,33,38]. Additionally, management was reluctant to both build new dams and restrict the height of existing dams, leading to a lack of quality control when it came to measuring the volume dumping rate at the Fundão dam [32]. Between 2012 and 2014, the dam was **certified safe and raised multiple times at a rate faster than that originally predicted** [31,53]. It was eventually found that the existing drainage system was structurally deficient, and to prevent cracking the crest of the dam had to be moved further upstream at the left abutment [31]. This is described in the "Dam Crest Readjustment" layer of Fig. 1. This adjustment represented an abandonment of the dam's original design and a **violation of its operating manual**, which rendered the system fundamentally different, and made results from prior stability tests invalid [32,38]. The

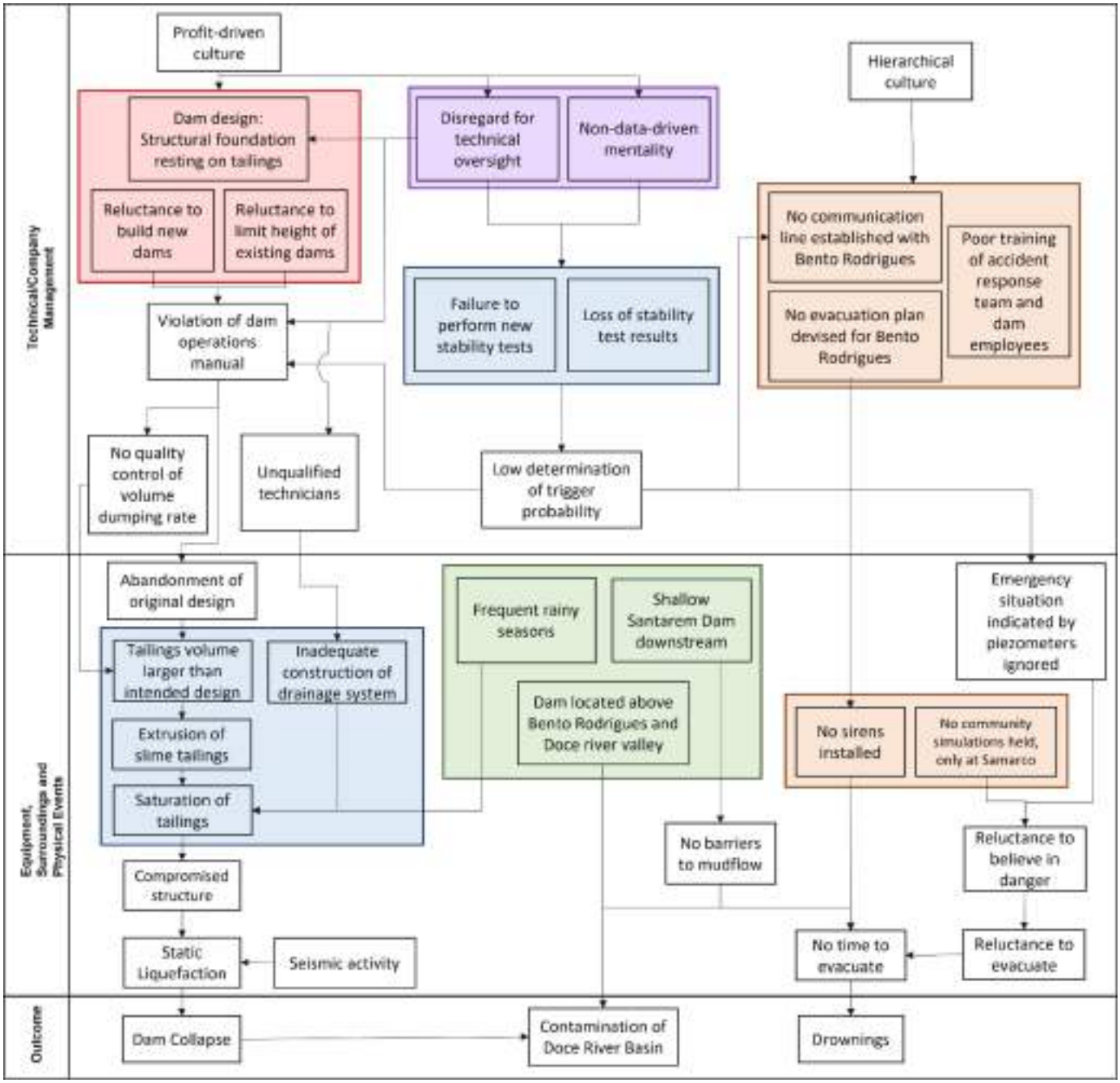

Fig. 5. Partial AcciMap for the Mariana Dam Disaster.

adjustment also had the effect of reducing the dam "beach length" to below its required 200 m, at one point reaching less than 60 m from the crest, which facilitated the extrusion of slime tailings underneath the dam structure and accelerated liquefaction [53]. A hierarchical organization at the facility along with poor communication across its different levels meant that these findings did not flow smoothly between the board of directors, company management, dam employees and the accident response team at the site [32]. Employees working for companies subcontracted by Samarco were **not trained in dam collapse emergency response** [53]. The proposed communication line with the leadership of Bento Rodrigues was not established even after structural and drainage deficiencies were identified, and thus no publicly available evacuation plan for the community was devised [62].

To understand how this was allowed to occur, the topmost two rows of the AcciMap are examined in Fig. 6. These layers deal with issues of regulation, inspection, government, and overall societal environment.

Ultimately, the disregard for technical oversight and the lack of a data-driven mindset was compounded by major conflicts of interest [40] with the organizations which certified the safety of tailings dams in Brazil. First, a lack of qualified inspection technicians working for the Brazilian government [56] allowed Vale and Samarco the right to "self-monitor", hiring external consulting firms to conduct the yearly safety audits [40,50,63]. One such firm provided technical reports highlighting the risks of each modification of the dam during its lifespan, and it provided recommendations for how to mitigate these risks, the majority of which were ignored by subsequent consulting firms which issued the multiple safety certifications the dam was awarded between 2011 and 2015 [32]. The ability for dam management to **constantly replace their external auditors** allowed for safety-critical information to be lost, buried, or not properly communicated during these transition periods. Additionally, the lack of legal frameworks governing tailings dam design, which would have enforced the use of best practices and best available technology [64], allowed for a shoddy, cost-efficient technique to prevail regardless of its hazards, which facilitated the aforementioned design and operational flaws of the Fundão dam from being approved and disregarded in the first place. An **inadequate accident management system** at the time also meant that even if these flaws had been identified, no regulatory or oversight agency would have had the information or power to mandate changes to the dam's design, construction and operation [37,55].

4. Brumadinho dam disaster analysis

4.1. Accident timeline

Dam I at the Córrego do Feijão iron mine in Brumadinho, Minas Gerais, was a tailings dam operated directly by Vale S.A., the parent company of Samarco. It was designed and constructed in the same upstream manner, but it was aimed primarily at storing sand tailings only [15]. This waste was vertically accumulated through successive uphill deposition, until the dam was eventually decommissioned in 2016 and management's main focus turned to the costly maintenance of the deposited tailings. Between 2016 and 2019, water was allowed to accumulate over the dam surface, which increased seepage erosion and chronically weakened the structure, and this is ultimately believed to have led to the collapse [34,36].

This accident timeline is adapted from the one presented in the official accident report by an independent consulting committee [35], as well as additional studies on the accident [46,47]. Without a confirmed trigger, at 12:28 PM on January 25, 2019, Dam I suffered a catastrophic failure, releasing approximately 9.7 million cubic meters in a wave with a 30-meter crest. Seismic records indicated low amplitude ground vibrations which began approximately 28 s before the first deformations were detected on the dam, but these were ruled to have been a result of the imminent collapse, rather than its cause. The failure initiated near the central part of the dam crest, followed by a noticeable bulge at the toe near the left abutment. Five seconds later, turbid water erupted at the foot of the dam near the right abutment, followed by a complete failure extending across the entire length of the dam toe, which released the mudslide that quickly reached the mine facility. Each layer of tailings then successively slipped and dislodged the next layer forward, resulting in a nearly vertical wall of natural ground being left over where the dam originally stood. The mudslide, traveling at about 120 km/h, first touched the foot of the adjacent Dam VI before hitting the mine's loading station, administrative area, and two sediment retention basins. Sirens had been installed but not yet tested across the entire facility, and none were activated. The administrative area included the control center for all alert systems, as well as the main cafeteria, where most of the dam's workers were having lunch. This resulted in a much higher death toll than the accident in Mariana, capping at 259 confirmed deaths with 11 people still missing. The tailings wave then traveled down the valley engulfing a hotel, several rural properties, a railway bridge, and 132 railway cars, before reaching the Paraopeba River at 3:50 PM [46], where it came to a rest. The contamination had a significant effect on the river ecosystem, and hindered access to drinking water along all adjacent settlements, as well as the large neighboring city of Belo Horizonte. This sequence of events is condensed in Table 2, and the relative locations of the dam and its facilities are shown in Fig. 7.

4.2. AcciMap analysis of the Brumadinho dam collapse

The inability to pinpoint a definitive trigger for the collapse on the day when it occurred should not detract from the fact that the Brumadinho accident, like Mariana, resulted from a complex causal basis that extended years prior to the accident. This depth and breadth of the accident pathogens and contributing factors are best understood by means of an AcciMap analysis, which is shown in its completed form in Fig. 8. The incremental assembly of the AcciMap is omitted here for practical reasons.

The bottom two rows represent the equipment, conditions and physical events which immediately led up to the accident, with the outcomes layer highlighting the three main events which constituted the accident and its consequences, namely: the dam collapse, the contamination of the Paraopeba river basin, and the death of 270 people. Similar to the Mariana AcciMap, three main categories of physical/environmental factors are identified, which involve the overall **location of the dam** (siting) relative to its facilities and surrounding inhabited areas, **the lack of drainage** and high phreatic level of the dam, and the lack **of a proper alert system**.

Having begun its construction in 1976, Dam I at the Córrego do Feijão mine had originally belonged to Ferteco Mineração S.A., which was purchased by Vale S.A. in 2001 [35,36]. Though the site had been inspected and approved prior to construction, regulations at the time were significantly looser, and the dam was allowed to be built despite directly disrupting the Córrego do Feijão watershed, which drained to the Paraopeba and later the São Francisco river valleys. At the time, few inhabited buildings were downstream of the dam, although the mine's processing plant, as well as the dam's main administrative facility and cafeteria were placed directly at the foot of the structure. Between 1976 and 2013, the dam was raised 10 times, [36] and the number of buildings erected downstream grew as well, creating a sizable accident pathogen in the event of a collapse.

Additionally, inspections dating back to 1995 had uncovered a high phreatic surface (water level) in the tailings deposits, which means **a large portion of the sandy tailings were not properly drained and were thus highly saturated** [35,43]. This was a particularly hazardous condition, which developed largely as a result of the lack of an internal drainage system for the starter dam. Most of the subsequent raisings included drainage systems, but the dikes themselves were made of low-permeability materials which held water within the tailings. To address the issue of the high phreatic surface, a system of deep horizontal drains (DHPs) was drilled into each of the dam's raisings in 2018.

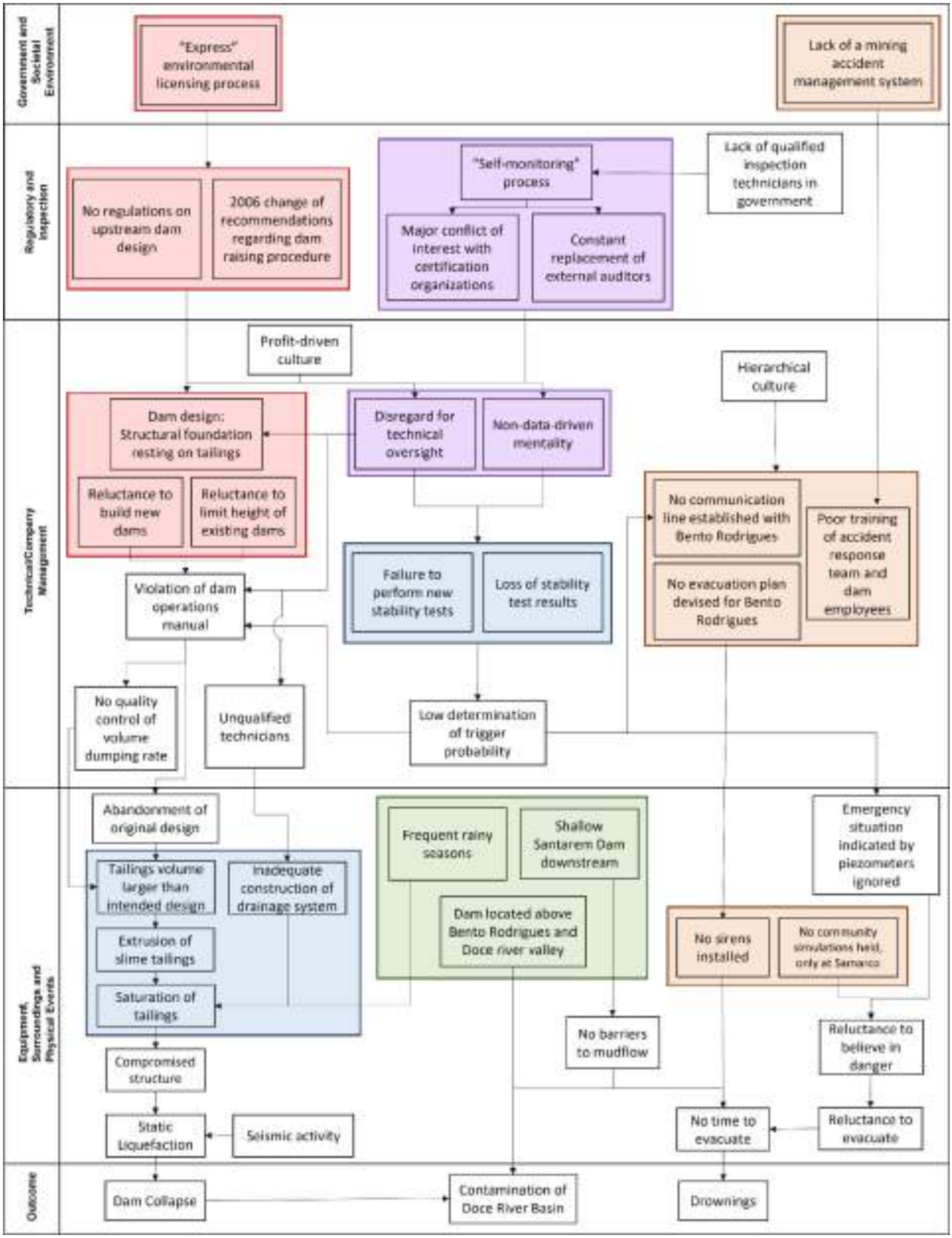

Fig. 6. Complete AcciMap for the Mariana Dam Disaster.

Table 2
Condensed Timeline of Brumadinho Dam Disaster.

1995	Dam risk to neighboring communities addressed, no evacuation plan created
2005	Study conducted on dam's susceptibility to liquefaction
2008–2013	Large seepages detected on dam's side
2014	Dam was certified as "safe", raised several times
2016	New liquefaction test conducted by Vale
January 25, 2019	**Day of the Accident**
12:27 PM	Low amplitude ground vibrations detected on dam
12:28:21 PM	Failure initiated near the central part of the dam crest, beginning with turbid water erupting at the foot of the dam near the right abutment, followed by complete failure extending across the entire length of the dam toe
12:28:40 PM	Dam structural debris reaches mine facilities
12:28:54 PM	Tailings contained within dam structure are released in a wave
3:50 PM	Wave of tailings reaches Paraopeba River

These consisted of perforated PVC tubes which gradually extracted water from the sandy tailings and redirected it through a network of canals to the riverbed at the bottom of the valley. They were selected for their low cost, despite general consensus in the scientific community of their inefficiency in lowering the phreatic surface in the short term [35, 52]. A major erosion failure during the installation of DHP 15 in June 2018 heralded the failures in accident response that were to come, as **no alarm systems activated**, and the emergency procedure devised by Vale for this **accident scenario had not been communicated to all workers at the dam**.

In 2017, in the wake of the Mariana disaster, an ordinance was issued by the National Department for Mineral Production (DNPM) requiring that sirens be installed at all iron ore dams and adjoining communities located in the "self-rescue zone", with a deadline of June 2019 [35]. The 14 sirens at Dam I had been installed by October 2018, but the testing of the system was scheduled for early 2019 [35]. The only external accident drill conducted by Vale in the communities of Córrego do Feijão and Vila Ferteco occurred before this, in August 2018, and was focused on the collapse of the neighboring Menezes I dam, not the dam which ultimately failed in January 2019 [36]. During this drill, temporary evacuation route signage was installed in the communities, and air horns were utilized in place of the real sirens, which had not yet been installed [36]. Not only were **residents unaware of the emergency notification system's real sound**, but many were unaware of even living in a potentially affected area [36]. **On the day of the collapse no sirens were activated** [53]. The sirens at the dam facility itself were immediately buried during the collapse, and the siren activation system for downstream communities was not directly linked to the dam monitoring, it was instead managed by a control center (CECOM) located offsite, which needed to be informed of a state of emergency prior to activation of sound systems [36]. Even had the alarm systems been linked to the dam monitoring system, the sensors had been found to produce discrepant data mere days before the collapse [44].

The next two rows of the AcciMap represent the technical and company management elements of the failure. As illustrated in the

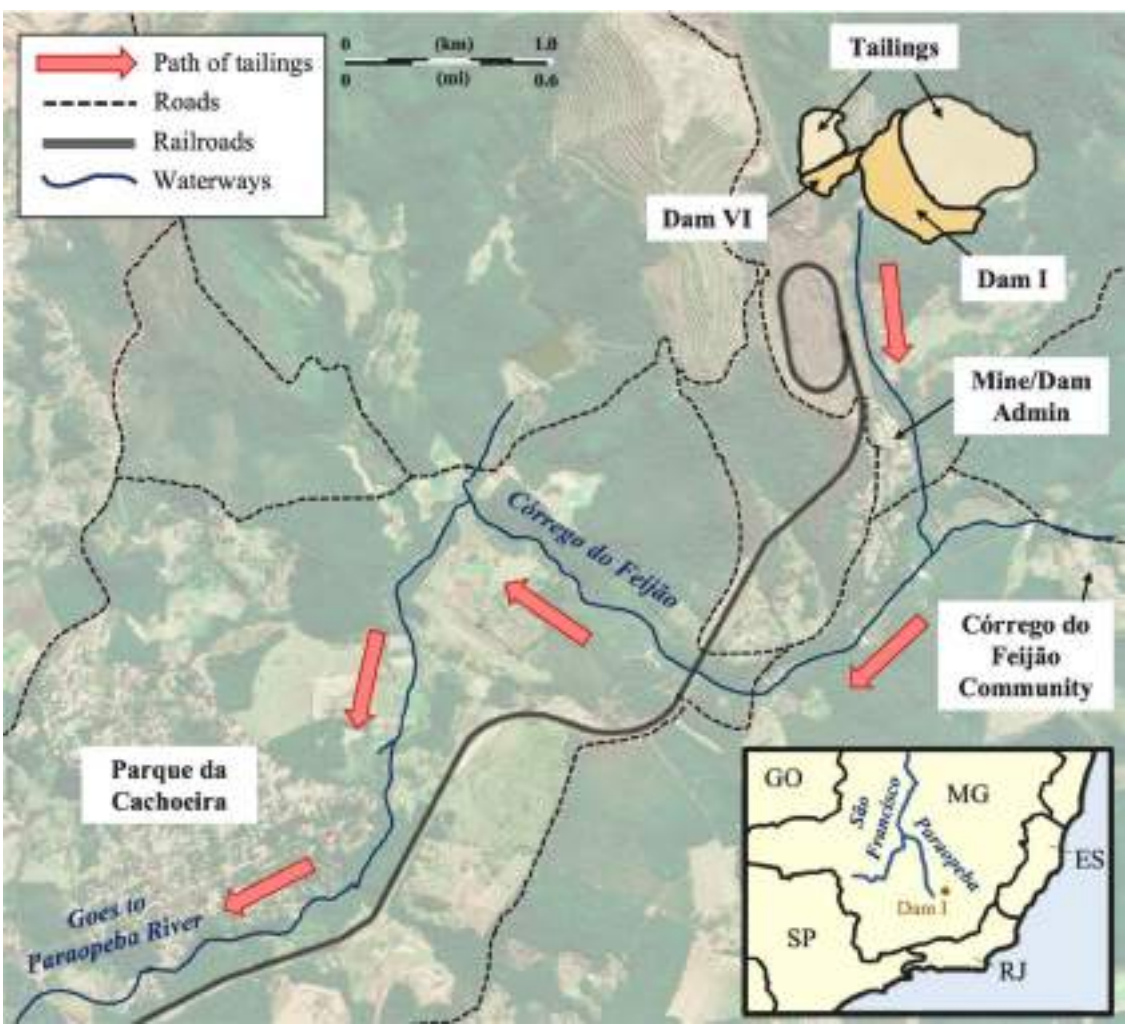

Fig. 7. Annotated satellite Image of Dam I and the surrounding area prior to the accident (adapted from Google Earth).

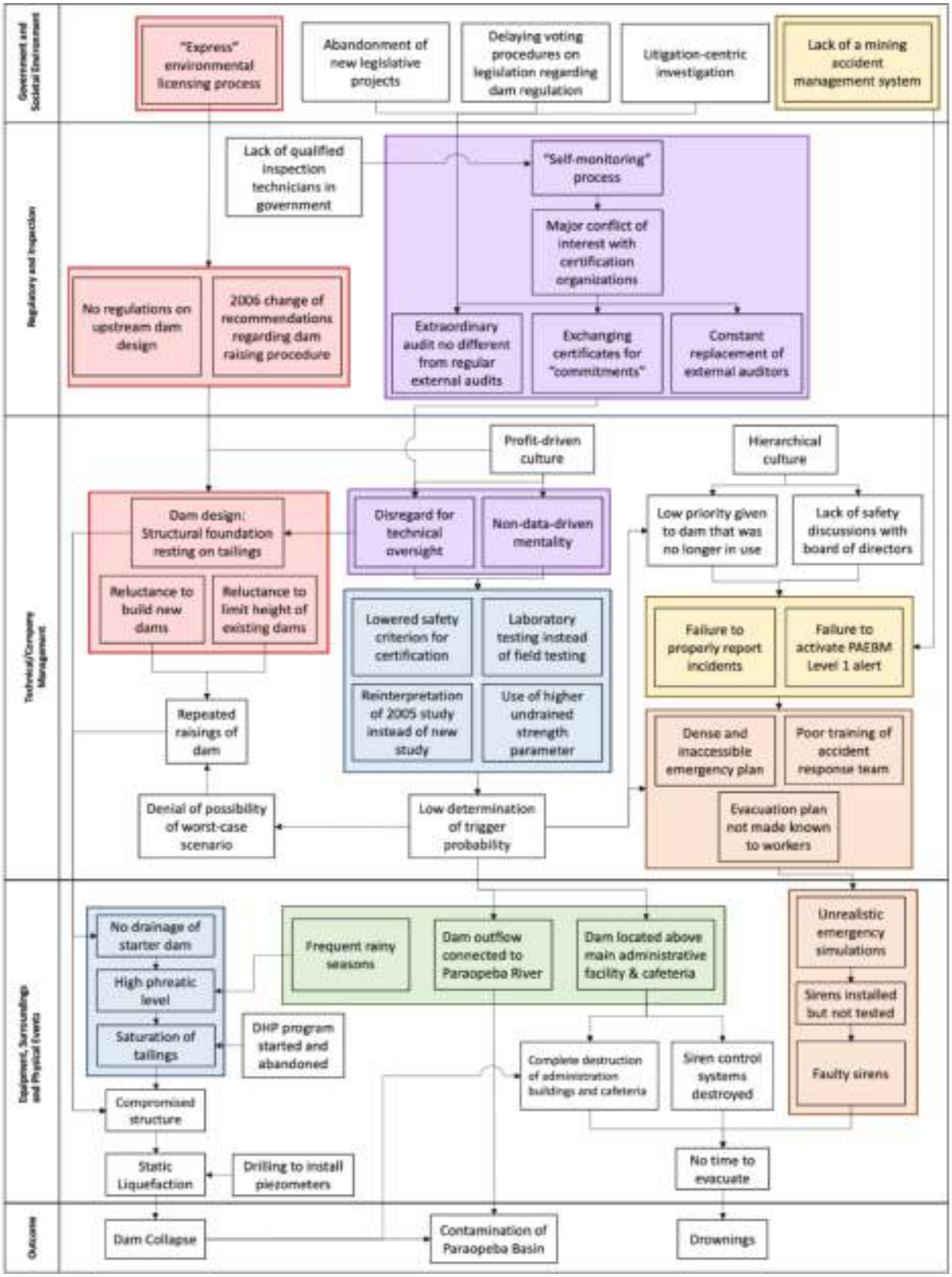

Fig. 8. Complete AcciMap for the Brumadinho Dam Disaster.

figure, the accident pathogens which allowed for the dam's precarious placement and lack of maintenance stemmed from a low determination of trigger probability. Studies following the 1995 inspection which uncovered a high phreatic level obtained similar results, and a study of the dam's susceptibility to liquefaction was conducted in 2005 [35]. Between 2008 and 2013, the dam was raised twice and the consulting company that had provided safety recommendations to the Mariana dam's management recommended a new liquefaction study each year [35,53]. In a major flaw in observability-in-depth, the auditors who performed this study in 2014 **did not collect any new liquefaction data, choosing instead to reevaluate the results obtained in 2005 with new methods** [35]**.** The two additional raisings had turned the dam into a fundamentally different structure, and the results of this reexamination study were deemed invalid by a future investigation panel [35]**.** Upon repeated requests by the auditors, a new liquefaction test campaign was commissioned in 2016 by Vale, who subcontracted it to two other consulting firms. The methodology employed in this study yielded safety metrics significantly lower than previously estimated, but the dam's **safety certification was issued** that same year, following a new test campaign with a revised methodology. This was achieved **by lowering the overall safety criterion for certification**, as well as the use of a higher undrained strength parameter than the dam truly possessed [35].

This disregard for technical oversight was fueled by the same cost-driven mentality which plagued Samarco prior to the Mariana collapse. Additionally, the deactivation of Dam I in 2016 placed it low on the company's priority list, and its safety condition was not further discussed with the company's board of directors. This lax reporting culture culminated in the failure to activate the Emergency Action Plan for Mining Dams (PAEBM) following the aforementioned incident with DHP 15. This action plan, which was defined individually for each dam in accordance with the DNPM in 2017, consisted of a set of action items based on a hazard scale ranging from level 1 (potential for compromised safety) to level 3 (imminent or immediate collapse) [36,65]. However, the current system of PAEBMs has been shown to be ineffective in providing response plans beyond certain specific categories of failures [55]. By limiting itself to considering only "states of emergency", the system fails to prevent the initiation of accident sequences, and instead focuses on their de-escalation [55]. Additionally, it has been shown to be dense and inaccessible to a majority of actors in the tailings dam safety value chain, as the different PAEBM levels of alert can only be activated by company management [36]. Ultimately, the reason for the non-activation of this plan during the DHP 15 incident was cited in an internal report by Vale's geotechnical team [35] as a **lack of preparedness to declare a state of emergency and activate the relief plan**. As a result, most dam workers were oblivious to the specifics of the dam's emergency plans, and the emergency drills that followed failed to reflect the gravity of a potential collapse.

The final two layers of the AcciMap deal with issues of regulation, inspection, government, and overall societal environment, and they share many of the issues that plagued the Mariana facility discussed earlier. These include **major conflicts of interest in the self-monitoring process** [40], and an overall **lack of regulations on upstream dam design** [37,55,64]**.** A notable change which occurred as a result of the 2015 collapse was the ordering of an "Extraordinary Technical Audit" to further inspect all tailings dams in Minas Gerais. However, this procedure was still being undertaken by consulting firms subcontracted by Vale, rather than by government-appointed auditors, which failed to resolve the aforementioned conflicts of interest. Additionally, **dam safety certifications issued by consulting firms were found to have been exchanged for "commitments" to improving safety measures** [35,50], rather than prior improvements that had already been executed. Each of these issues stems from the same governmental flaws outlined earlier, along with a chronic inefficiency of the Brazilian legal system in addressing the issues that had already been identified. This allowed for new legislative projects, such as PSL No. 224 in 2016 which had the objective of restructuring and reinforcing the effectiveness of the National Policy for Dam Security [66], to be either shelved, never voted upon, or deemed already resolved without any further action.

5. Discussion

This section provides a synthesis and discussion of the findings in the previous dam collapse timelines and AcciMap analyses. Section 5.1 includes a discussion of the main failures during the investigation of, and response to the Mariana accident, which played a role in the development of similar accident pathogens in Brumadinho or failed to prevent it. This is followed in Section 5.2 by a set of recommendations for improving the safety of tailings dams and enhancing the accident investigation process.

5.1. Limitations in accident investigation and (regulatory) response

Following the discussion of these two successive accidents, it is important to examine the reasons for the shortcomings in the accident investigation into the Mariana dam collapse and the (regulatory) response to prevent the Brumadinho accident from occurring. With the Mariana accident, multiple investigations concluded that the accident occurred primarily due to the actions of Vale S.A., the parent company of Samarco, which ran the Fundão dam. The main report [31] investigating the collapse placed strong emphasis on the degree of negligence that Vale displayed leading up to the accident. Design flaws in the original dam structure were compounded with the ignoring of warnings given both by instrumentation placed throughout the dam's walls and by external auditors. The fact that dam employees and much of management alike were unaware of the heightened hazard levels and were ill-prepared when it came to considering various accident scenarios at the dam are distinct organizational factors that contributed to the accident (poor communication, flawed risk management). These pathogens, when triggered and compounded with the lack of emergency preparedness, greatly accentuated the consequences of the dam collapse, placing at risk not just the lives of the company's employees, but also that of the residents in the surrounding communities [13,42].

While the reports presented a comprehensive investigation of the engineering and technical causes of the accident, their main flaw was in their failure to examine organizational and regulatory factors to the accident [26,45], and more generally contributing factors captured in the upper layers of AcciMap when investigating accidents (see Figs. 6 and 8 for details). While identifying the sequences of events leading to an accident and the underlying technical causes is necessary in an investigation, by focusing only on technical causes, investigators miss a significant number of contributing factors, organizational and regulatory, for example, which allowed many accident pathogens to develop in the years prior to the collapse and ultimately contributed to the accident [26]. As mentioned earlier, the "Report on the Immediate Causes of the Failure of the Fundão Dam" [31] conducted an extensive investigation of the technical factors at Mariana, but there was no equivalent report detailing these other non-technical factors [45]. The main report [32], which did consider such aspects, was drafted for litigation purposes only, aimed at determining the culpability of specific individuals and firms, rather than preventing the recurrence of these factors in the future. This nuance is subtle but important. Instead of recommending or mandating proper communication protocols with communities adjacent to tailings dams in the event of an accident, for example, or requiring clear policies for risk management and emergency preparedness, or imposing independent inspection and certification, assigning blame for technical errors leaves many of accident pathogens in place and fails to learn and act on the important lessons that can prevent recurrence of similar accidents [26]. This obfuscation of the organizational, regulatory and social aspects of the accident [41] beyond the litigation procedures highlights a potential limitation in the safety literacy of

Brazilian investigators and their agencies. The conflicts of interest, which pervaded many of the inspections prior to the collapse, are seen again in the post-accident investigation and response, as the new "extraordinary audit" mandated at every dam was once again permitted to be a self-reported effort. This was a lesson neither learned nor properly identified. Alternatively, this maintenance of the status quo might be an indication of the investigative authorities unwillingness or incapability of challenging the underlying structure for safety oversight, likely a result of the economic and social influence held by large corporations in Brazil. Ultimately, the identification and implementation of real change to the companies' and regulatory agencies' safety structure and oversight was forfeited in favor of superficial risk management and cosmetic transformations in which conveying an image of safety improvement appears more important than actually enacting lasting change to meaningfully improve safety [38,52,54].

In short, the investigations into the Mariana dam collapse were lacking in three ways: (1) they focused on technical factors but failed to expand the same effort into other soft contributing factors (e.g., organizational and regulatory) within and beyond the organization that oversaw the dam; (2) they failed to identify an important lesson or were unwilling to challenge the existing status quo of self-reported safety audits and compromised (not independent) inspections [26]; and (3) they heavy-handedly pursued litigation at the detriment of identifying and implementing meaningful safety improvements. These flaws kept many of the accident pathogens in place and primed for being triggered a few years later at the Brumadinho dam.

Fortunately, following the Brumadinho accident, an independent auditing committee known as the CIAEA [34,35] committee was formed, and was mandated with identifying the broader causal factors in that collapse and issuing recommendations. Expanding upon the structure of the litigation report of the Mariana collapse, the CIAEA conducted a more encompassing investigation of the Brumadinho accident. The report contained a section dedicated to "Recommendations", which specifically targeted non-technical causal factors [35], which were missing in a majority of the articles written following the Mariana accident [31,45,54]. Taking the CIAEA findings into consideration, the following section presents further recommendations towards preventing tailings dam collapses in the future.

5.2. Recommendations

In light of the shortcomings identified previously, this section presents recommendations for the improvement of safety conditions in tailings dams across Brazil and the world. These can be broken down into three broad categories, namely (1) procedural/design recommendations for the operation of tailings dams, (2) recommendations for improving the safety culture in the Brazilian government and legislative system, and (3) recommendations regarding the methods by which tailings dam failures should be investigated and addressed. These recommendations have been generated by the authors through a synthesis and expansion of those proposed in the existing literature. The themes present here, such as more effective monitoring, inspection and auditing of facilities, safety-driven evaluations of changes to the original design, and better contingency plans in case of failure are consistent with those proposed in similar works, such as [67].

5.2.1. Improving the safety of tailings dam design and operation

Five primary recommendations are presented here with regards to the safe design and operation of tailings dams. Such recommendations act both to prevent accidents and to mitigate their consequences in the event of a trigger. They are:

(a) **Facility siting:** Ideally, **all buildings should be placed far enough away from the toe of the tailings dam**, so that every individual in the building is capable of being notified of a failure and evacuating before the building encounters any moving fluids. Drawing from the aviation industry's standard 90 s rule for the emergency evacuation of aircraft, a similar rule can be devised such that the time required for the dam's deposit to reach the building assuming a complete dam collapse must be greater than the combined time for the failure to be detected by a facility's alert systems and informed to those at risk, and for every such individual to evacuate the building and arrive at a safe location. This can be achieved via the relocation of facilities to sites further downstream from the dam, or the installation of flood control measures such as spillways or flood walls to reduce the speed and volume of moving dam deposits.

(b) **Alerting systems:** There is to be a **time-effective alerting system** at all locations subject to damage from a tailings dam collapse. The system must be robust and de-localized, so that a failure at a single point in the transmission of the signal does not result in the entirety of the message being lost. Additionally, the system must be directly linked to the dam monitoring instrumentation, so that it is not reliant on an off-site control center being notified of a failure before sirens are activated. [48–50]

(c) **Community drills:** Focus should be placed on the development and implementation of effective **simulation campaigns in the communities** surrounding large tailings dams, with the purpose of training the population to coordinate the evacuation of their homes and workplaces. This campaign should include dam collapse drills representative of real emergency scenarios, as well as the publicization of maps and evacuation procedures which were, for the most part, either kept private by Vale and Samarco or made temporarily available during the limited simulations carried out in Brumadinho in 2018. Participation in such educational campaigns should come at no cost to the population.

(d) **Road access:** There must be an **improvement of road access to potentially afflicted regions**. The search and rescue process following the Mariana disaster was rendered extremely difficult as a result of poor road infrastructure, forcing much of the operation to be conducted by helicopter. Action should be taken by the local or federal government to improve access to said communities for the delivery of aid in the event of a major dam collapse.

(e) **Change management programs:** The implementation of **change management programs** at tailings dam facilities is crucial to understanding the safety implications of proposed modifications [68–70]. Major changes to the physical structure or operational procedures in tailings dams, including successive raisings, should only be authorized after a comprehensive data-driven re-evaluation of dam safety by independent contractors. This re-evaluation must include measurements of key dam safety parameters, as well as a plan for continuous monitoring of the system's response to the proposed changes [12]. The results of this re-evaluation should be subject to scrutiny by public authorities. This is particularly important in the case where changes would represent a violation of dam safety manuals, as was the case with the dam crest readjustment at the Fundão Dam [32,38].

5.2.2. Improving the safety culture of Brazilian government and legislation

The recommendations proposed here aim to set an example for improving the safety culture within the Brazilian government and legislative system by restructuring the chains of accountability in Brazilian mining operations. These are:

(a) **Reevaluation of chains of inspection:** There should be a **reevaluation of the status quo with regards to the chains of inspection** at tailings dams in Brazil. The practice of self-monitoring proved a contributing factor to the Mariana collapse, and the refusal to challenge the existing system allowed its pathogens to be triggered again in the Brumadinho accident.

This policy should be eliminated, so as to prevent conflicts of interest like financial or social incentives for investigators to produce results of any particular nature. Government at the federal, state, and local levels should embrace their responsibilities to regulate, oversee, and monitor any dangerous activities. **Federal and state governments should create specific agencies** run by trained technicians to oversee and authorize major changes to tailing dams. One potential approach to this, which recognizes the logistical challenges faced by the Brazilian government due to the low supply of qualified tailings dam inspectors [56] in the public sector, is the creation of expert panels hired directly by governmental authorities to conduct independent investigations [26], but funded by mining companies themselves [52,54]. Additionally, **local governments should demand that emergency and evacuation plans** be put in place in facilities and their surrounding communities. City councils must have the ability to oversee and approve such plans both before dam operations begin, and during all stages of the life cycle shown in Fig. 1.

(b) **Restructuring of chains of accountability:** There should be a restructuring of the chains of accountability at tailings dams by **leveraging regulatory authorities' power to approve and revoke licenses of operation**. Licenses of operation for dangerous facilities should be revised on a yearly basis, and only issued upon presentation of safety and risk management plans, which must demonstrate compliance with both technical and social risk acceptability criteria [71]. These criteria should consider not only the risk within the facility itself, but it's *situated* social risk, which combines the risk of the structure with the inherent vulnerabilities of the local context in which it is constructed [8]. Additionally, licensing should be contingent on full payment of any open environmental fines, civil/criminal fines and compensation issued by confirmed court decisions. This places safety and risk management in the best interest of the companies operating tailings dams, and incentivizes safety at both the company management and governmental level.

5.2.3. *Improving tailings dam accident investigation*

The recommendations for the improvement of tailings dam accident investigation can be built on from the discussion of the specific shortcomings in the Mariana dam disaster investigation covered in Section 5.1. This section will focus on the restructuring of the chains of oversight before and after accidents at Brazilian tailings dams. The recommendations are:

(a) **Examination of High-Level Contributing Factors:** Investigation reports should have a section focusing on **non-technical contributing factors to tailings dam accidents**. Most of the points highlighted in the upper levels of the Mariana AcciMap were seen again in the Brumadinho AcciMap, and thus had been known to Vale and Samarco prior to the second collapse. These should have been addressed in the official reports on the causes of the Mariana accident, not solely during the litigation procedure. Investigation reports should examine contributing factors from company management, regulatory bodies, and overall societal culture, and legislative action [37] should be undertaken to preclude them, rather than only to punish individuals and firms.

(b) **Implementation of Safety Management Systems:** The **implementation of interconnected safety/near-miss management systems** [37,55] is of paramount importance to continuously feed learning loops and expand the imagination of failure in tailings dam operations. These systems should operate at both the company level, as well as the national level to guide future regulation. The aforementioned PAEBM system must be expanded to incorporate the collection and processing of data regarding all minor incidents, breaches of protocol, and general heightened hazard levels, to bring the safety management strategy all the way to the initiating event level and minimize the formation of "states of emergency" in the first place.

6. Conclusion

The objective of this paper was to conduct a critical analysis of the technical, managerial and governmental causal factors which allowed for the collapse of two tailings dams in Mariana and Brumadinho, Minas Gerais, Brazil, with particular emphasis on the failures in the investigations to these accidents. Each accident was described in terms of its timeline, and the key connections between different factors which were compounded to cause failure were incrementally modeled through an AcciMap approach. These AcciMaps were then compared and contrasted to identify which key components of the Mariana accident went unaddressed during the subsequent investigation, and a set of recommendations was provided to adequately resolve some of these issues and prevent their recurrence.

Overall, the results of this study highlighted a major insufficiency in the imagination of failure on the part of Samarco and Vale S.A., driven largely by a profit-centric mindset which prompted numerous inspections and recommendations to be largely ignored. This was facilitated by an overarching governmental structure which allowed for the self-monitoring of large-sale mine operations, creating major conflicts of interest when best practices in safety management were undermined by financial incentives. Ultimately, the lack of focus placed on the non-technical aspects of the Mariana accident when it came to developing safety recommendations, came at the cost of hundreds of lives in Brumadinho, with many of the managerial and governmental issues observed in one accident being present in much the same way in the other. Since the Brumadinho accident, a number of measures have already been taken in line with those suggested here, but without a further restructuring of the chains of responsibility and accountability in Brazilian mining operations as a whole, the country will be forced to continue bearing the social and environmental costs of major accidents.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Rodrigo L. Rose: Methodology, Formal analysis, Writing – review & editing, Conceptualization, Writing – original draft, Visualization. **Sohan R. Mugi:** Methodology, Formal analysis, Writing – review & editing, Conceptualization, Writing – original draft, Visualization. **Joseph Homer Saleh:** Conceptualization, Formal analysis, Methodology, Writing – review & editing, Visualization, Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

The authors would like to dedicate this paper to the memory of co-author Joseph Homer Saleh, whose contributions to this work and to

the field of system safety are invaluable. The authors would also like to acknowledge the support and guidance provided by Dr. Karen Feigh during the review and editing process, as well as the work of Achal Kikkeri toward the initial manuscript.

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